Institution: National Higher School of Mathematics
Course: Image Analysis WORKSHOP
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Medical Image Denoising: BM3D, CNN, and Swin Transformer Comparison

PHASE 1: Preparation

Loading the Dataset and Adding Different Types of Noise

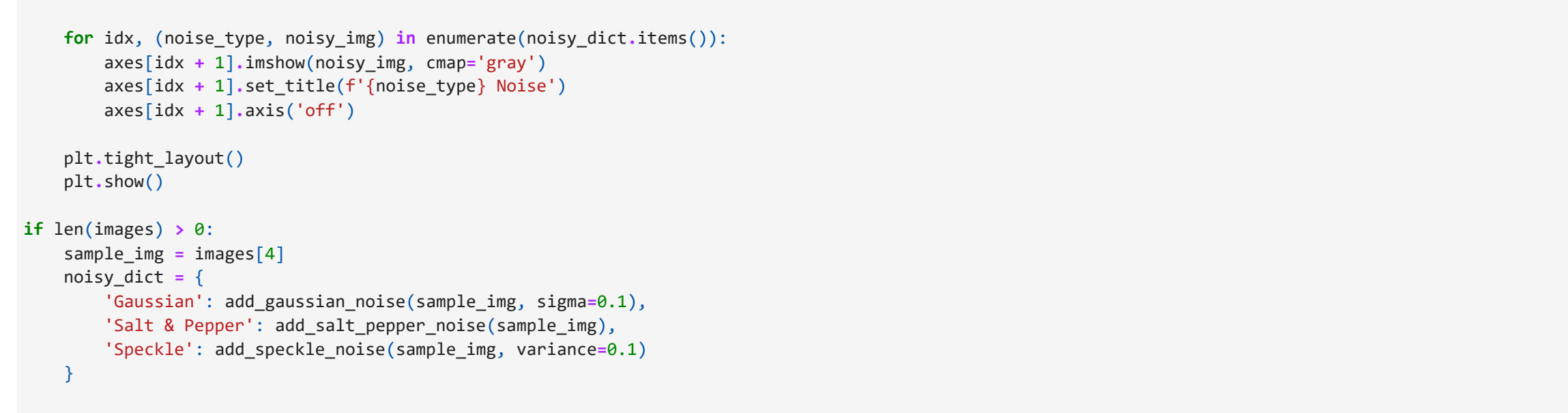
In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
import cv2
import os
from skimage import io, img_as_float, color
from skimage.metrics import peak_signal_noise_ratio as psnr
from skimage.metrics import structural_similarity as ssim
import random
import warnings
warnings.filterwarnings('ignore')
from scipy import ndimage
from skimage import io, color
import torch
import tensorflow as tf
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
from sklearn.model_selection import train_test_split
import time

# Check if CUDA is available
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Using device: {device}")

# Set random seeds for reproducibility
def set_seed(seed=42):
    random.seed(seed)
    np.random.seed(seed)
    torch.manual_seed(seed)
    if torch.cuda.is_available():
        torch.cuda.manual_seed(seed)
        torch.cuda.manual_seed_all(seed)
        torch.backends.cudnn.deterministic = True
        torch.backends.cudnn.benchmark = False

set_seed(42)
```

```
Using device: cpu
```

In [2]:
```
def load_dataset(dataset_path):
    images = []
    image_paths = []
    extensions = ['.jpg', '.jpeg', '.png', '.bmp', '.tif', '.tiff']

    for root, dirs, files in os.walk(dataset_path):
        for file in files:
            if any(file.lower().endswith(ext) for ext in extensions):
                try:
                    img_path = os.path.join(root, file)
                    img = io.imread(img_path)
                    if len(img.shape) == 3:
                        img = color.rgb2gray(img)
                    img = img_as_float(img)
                    if img.shape[0] != 256 or img.shape[1] != 256:
                        img = cv2.resize(img, (256, 256))
                    images.append(img)
                    image_paths.append(img_path)
                except Exception as e:
                    print(f"Error loading {file}: {e}")

    print(f"Loaded {len(images)} images")
    return images, image_paths

dataset_path = 'Dataset'
images, image_paths = load_dataset(dataset_path)
```

```
Loaded 120 images
```

In [3]:
```
def add_gaussian_noise(image, mean=0, sigma=0.1):
    noise = np.random.normal(mean, sigma, image.shape)
    noisy = image + noise
    return np.clip(noisy, 0, 1)

def add_salt_pepper_noise(image, prob=0.1):
    noisy = image.copy()
    salt_pepper = np.random.random(image.shape)
    noisy[salt_pepper < prob/2] = 1
    noisy[salt_pepper > 1 - prob/2] = 0
    return noisy

def add_speckle_noise(image, variance=0.1):
    noise = np.random.randn(*image.shape) * np.sqrt(variance)
    noisy = image + image * noise
    return np.clip(noisy, 0, 1)

# Display noisy images
def display_noisy_images(original, noisy_dict):
    fig, axes = plt.subplots(1, 4, figsize=(20, 20))

    axes[0].imshow(original, cmap='gray')
    axes[0].set_title('Original Image')
    axes[0].axis('off')

    for idx, (noise_type, noisy_img) in enumerate(noisy_dict.items()):
        axes[idx + 1].imshow(noisy_img, cmap='gray')
        axes[idx + 1].set_title(f'{noise_type} Noise')
        axes[idx + 1].axis('off')

    plt.tight_layout()
    plt.show()

if len(images) > 0:
    sample_img = images[4]
    noisy_dict = {
        'Gaussian': add_gaussian_noise(sample_img, sigma=0.1),
        'Salt & Pepper': add_salt_pepper_noise(sample_img),
        'Speckle': add_speckle_noise(sample_img, variance=0.1)
    }

    display_noisy_images(sample_img, noisy_dict)
```



PHASE 2: Denoising the images using BM3D and CNN

Denoising using BM3D

In [4]:
```
import bm3d

def denoise_bm3d(noisy_image, sigma_psd=0.1):
    denoised = bm3d.bm3d(noisy_image, sigma_psd=sigma_psd)
    return denoised
```

Denoising using CNN

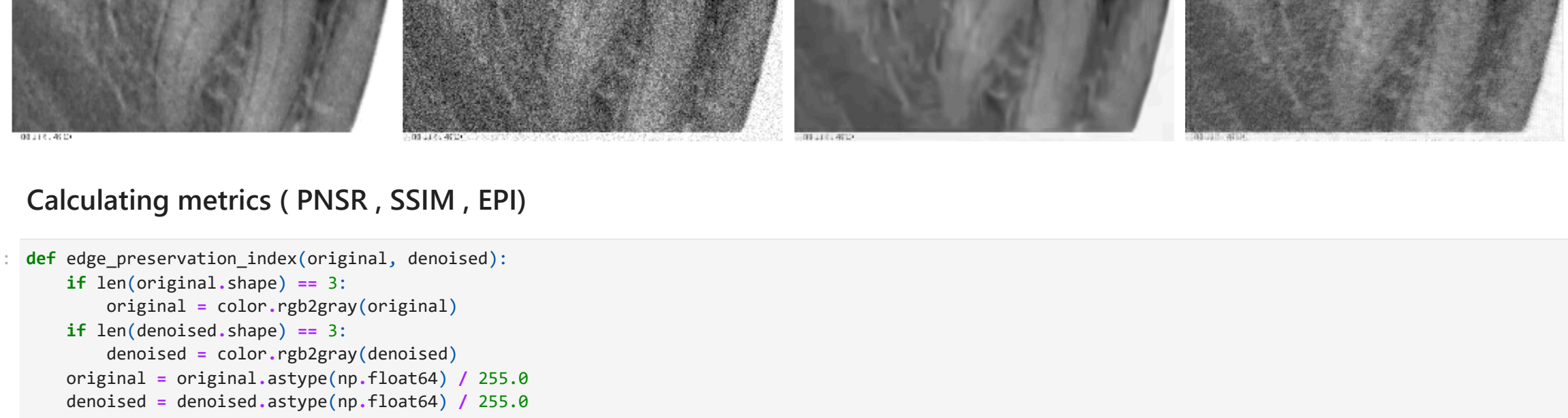
In [5]:
```
# Optimized CNN Denoising Model for small dataset
class SimpleDenoisingCNN(nn.Module):
    def __init__(self):
        super(SimpleDenoisingCNN, self).__init__()

        # Further reduced channels for 120 images
        self.enc1 = nn.Conv2d(1, 16, kernel_size=3, padding=1)
        self.bn1 = nn.BatchNorm2d(16)
        self.enc2 = nn.Conv2d(16, 32, kernel_size=3, padding=1)
        self.bn2 = nn.BatchNorm2d(32)

        # Decoder
        self.dec1 = nn.Conv2d(32, 16, kernel_size=3, padding=1)
        self.bn3 = nn.BatchNorm2d(16)
        self.dec2 = nn.Conv2d(16, 1, kernel_size=3, padding=1)

        self.relu = nn.ReLU(inplace=True)
        self.use_residual = True

    def forward(self, x):
        residual = x

        x1 = self.relu(self.bn1(self.enc1(x)))
        x2 = self.relu(self.bn2(self.enc2(x1)))

        x3 = self.relu(self.bn3(self.dec1(x2)))
        x4 = self.dec2(x3)

        if self.use_residual:
            x4 = x4 + residual

        return x4

    def count_parameters(self):
        return sum(p.numel() for p in self.parameters() if p.requires_grad)
```

In [6]:
```
class DenoisingDataset(Dataset):
    def __init__(self, images, noise_type='gaussian', noise_param=0.1):
        # Normalize images to [0, 1] range
        if images and images[0].max() > 1.0:
            images = [img.astype(np.float32) / 255.0 for img in images]

        self.images = [torch.FloatTensor(img).unsqueeze(0) for img in images]
        self.noise_type = noise_type
        self.noise_param = noise_param

    def __len__(self):
        return len(self.images)

    def __getitem__(self, idx):
        clean = self.images[idx]

        # Add noise efficiently
        if self.noise_type == 'gaussian':
            noise = torch.randn_like(clean) * self.noise_param
            noisy = clean + noise
        elif self.noise_type == 'salt_pepper':
            noisy = clean.clone()
            salt_pepper = torch.rand_like(clean)
            noisy[salt_pepper < 0.05] = 0.0
            noisy[salt_pepper > 0.95] = 1.0
        elif self.noise_type == 'speckle':
            noise = torch.randn_like(clean) * self.noise_param
            noisy = clean + clean * noise

        noisy = torch.clamp(noisy, 0.0, 1.0)

        return noisy, clean
```

In [7]:
```
def train_cnn_model_optimized(train_loader, model, epochs=10):
    criterion = nn.MSELoss()
    optimizer = optim.AdamW(model.parameters(), lr=0.001, weight_decay=1e-5)
    scheduler = optim.lr_scheduler.ReduceLROnPlateau(optimizer, mode='min', factor=0.5, patience=2)

    model.train()
    train_losses = []

    # Mixed precision training
    scaler = torch.cuda.amp.GradScaler() if torch.cuda.is_available() else None

    start_time = time.time()

    for epoch in range(epochs):
        epoch_loss = 0
        batch_count = 0

        for batch_idx, (noisy, clean) in enumerate(train_loader):
            noisy, clean = noisy.to(device), clean.to(device)

            optimizer.zero_grad()

            if scaler:  # Mixed precision training
                with torch.cuda.amp.autocast():
                    output = model(noisy)
                    loss = criterion(output, clean)

                scaler.scale(loss).backward()
                scaler.step(optimizer)
                scaler.update()
            else:
                output = model(noisy)
                loss = criterion(output, clean)
                loss.backward()
                optimizer.step()

            epoch_loss += loss.item()
            batch_count += 1

        avg_loss = epoch_loss / batch_count
        train_losses.append(avg_loss)
        scheduler.step(avg_loss)

        elapsed = time.time() - start_time
        print(f'Epoch {epoch+1}/{epochs}, Loss: {avg_loss:.6f}, Time: {elapsed:.2f}s')

    return train_losses

# Split data
train_images, val_images = train_test_split(images, test_size=0.2, random_state=42)

# Create optimized data loader
train_dataset = DenoisingDataset(train_images, noise_type='gaussian')
train_loader = DataLoader(
    train_dataset,
    batch_size=8,
    shuffle=True,
    num_workers=2 if torch.cuda.is_available() else 0,
    pin_memory=True if torch.cuda.is_available() else False
)

cnn_model = SimpleDenoisingCNN().to(device)
print(f"CNN Parameters: {cnn_model.count_parameters():,}")
train_losses = train_cnn_model_optimized(train_loader, cnn_model, epochs=5)
```

```
CNN Parameters: 9,697
Epoch 1/5, Loss: 0.047362, Time: 9.80s
Epoch 2/5, Loss: 0.008845, Time: 18.51s
Epoch 3/5, Loss: 0.004458, Time: 27.59s
Epoch 4/5, Loss: 0.003117, Time: 36.70s
Epoch 5/5, Loss: 0.002553, Time: 46.87s
```

In [8]:
```
# Test denoising methods
def test_denoising_methods(test_image, noise_type='gaussian'):
    if noise_type == 'gaussian':
        noisy = add_gaussian_noise(test_image, sigma=0.1)
    elif noise_type == 'salt_pepper':
        noisy = add_salt_pepper_noise(test_image)
    elif noise_type == 'speckle':
        noisy = add_speckle_noise(test_image, variance=0.1)

    # BM3D denoising
    bm3d_denoised = denoise_bm3d(noisy, sigma_psd=0.1)
    # CNN denoising
    cnn_model.eval()
    with torch.no_grad():
        noisy_tensor = torch.FloatTensor(noisy).unsqueeze(0).unsqueeze(0).to(device)
        cnn_denoised = cnn_model(noisy_tensor).cpu().squeeze().numpy()

    results = {
        'Noisy': noisy,
        'BM3D': np.clip(bm3d_denoised, 0, 1),
        'CNN': np.clip(cnn_denoised, 0, 1)
    }

    return results, test_image

# Test on a sample image
if len(val_images) > 0:
    test_img = val_images[0]
    results, original = test_denoising_methods(test_img, noise_type='gaussian')

    # Display results
    fig, axes = plt.subplots(1, 4, figsize=(20, 20))

    axes[0].imshow(original, cmap='gray')
    axes[0].set_title('Original')
    axes[0].axis('off')

    axes[1].imshow(results['Noisy'], cmap='gray')
    axes[1].set_title('Noisy')
    axes[1].axis('off')

    axes[2].imshow(results['BM3D'], cmap='gray')
    axes[2].set_title('BM3D')
    axes[2].axis('off')

    axes[3].imshow(results['CNN'], cmap='gray')
    axes[3].set_title('CNN')
    axes[3].axis('off')

    plt.tight_layout()
    plt.show()
```



Calculating metrics (PNSR , SSIM , EPI)

In [9]:
```
def edge_preservation_index(original, denoised):
    if len(original.shape) == 3:
        original = color.rgb2gray(original)
    if len(denoised.shape) == 3:
        denoised = color.rgb2gray(denoised)
    original = original.astype(np.float64) / 255.0
    denoised = denoised.astype(np.float64) / 255.0
    grad_x_orig = ndimage.sobel(original, axis=0)
    grad_y_orig = ndimage.sobel(original, axis=1)
    grad_x_den = ndimage.sobel(denoised, axis=0)
    grad_y_den = ndimage.sobel(denoised, axis=1)
    delta_I = np.sqrt(grad_x_orig**2 + grad_y_orig**2)
    delta_I_hat = np.sqrt(grad_x_den**2 + grad_y_den**2)
    mu_delta_I = np.mean(delta_I)
    mu_delta_I_hat = np.mean(delta_I_hat)
    dev_I = np.abs(delta_I - mu_delta_I)
    dev_I_hat = np.abs(delta_I_hat - mu_delta_I_hat)
    numerator = np.sum(dev_I * dev_I_hat)
    denominator = np.sqrt(np.sum(dev_I**2) * np.sum(dev_I_hat**2))
    if denominator == 0:
        return 1.0
    epi = numerator / denominator
    return np.clip(epi, 0, 1)


print(f"Edge Preservation Index (EPI) for BM3D: {edge_preservation_index(original, results['BM3D']):.4f}")
print(f"Peak Signal-to-Noise Ratio (PSNR) for BM3D: {psnr(original, results['BM3D']):.4f}")
print(f"Structural Similarity Index (SSIM) for BM3D: {ssim(original, results['BM3D'], data_range=original.max() - original.min()):.4f}")
print('-----------------------------------------------------')
print(f"Edge Preservation Index (EPI) for CNN: {edge_preservation_index(original, results['CNN']):.4f}")
print(f"Peak Signal-to-Noise Ratio (PSNR) for CNN: {psnr(original, results['CNN']):.4f}")
print(f"Structural Similarity Index (SSIM) for CNN: {ssim(original, results['CNN'], data_range=original.max() - original.min()):.4f}")
```

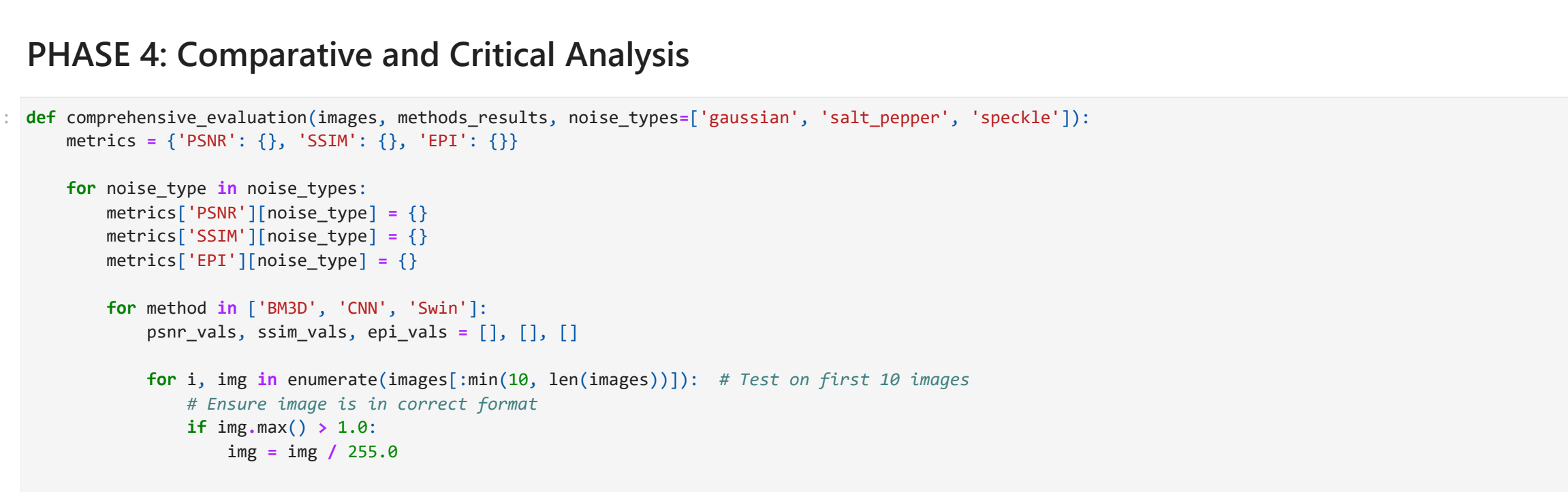
```
Edge Preservation Index (EPI) for BM3D: 0.9608
Peak Signal-to-Noise Ratio (PSNR) for BM3D: 30.8546
Structural Similarity Index (SSIM) for BM3D: 0.7776
-----------------------------------------------------
Edge Preservation Index (EPI) for CNN: 0.8769
Peak Signal-to-Noise Ratio (PSNR) for CNN: 25.9596
Structural Similarity Index (SSIM) for CNN: 0.5394
```

PHASE 3: Restoration Using Swin Transformer

In [10]:
```
IMG_SIZE = 128
DATASET_PATH = "Dataset"
BATCH_SIZE = 8
EPOCHS = 20

def load_images(folder):
    images = []
    for file in os.listdir(folder):
        path = os.path.join(folder, file)
        img = cv2.imread(path)
        if img is None:
            continue
        img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))
        img = img / 255.0
        images.append(img)
    return np.array(images)


clean_images = load_images(DATASET_PATH)
noisy_images = add_gaussian_noise(clean_images)
x_train, x_test, y_train, y_test = train_test_split(noisy_images,clean_images,test_size=0.2,random_state=42)


def swin_block(x, num_heads=4):
    dim = x.shape[-1]
    x_norm = tf.keras.layers.LayerNormalization()(x)
    attn = tf.keras.layers.MultiHeadAttention(num_heads=num_heads,key_dim=dim)(x_norm, x_norm)
    x = tf.keras.layers.Add()([x, attn])
    x_norm = tf.keras.layers.LayerNormalization()(x)
    mlp = tf.keras.layers.Dense(dim * 4, activation="gelu")(x_norm)
    mlp = tf.keras.layers.Dense(dim)(mlp)
    x = tf.keras.layers.Add()([x, mlp])
    return x


def patch_embedding(x, patch_size=4, embed_dim=64):
    x = tf.keras.layers.Conv2D(embed_dim,kernel_size=patch_size,strides=patch_size,padding="same")(x)
    shape = x.shape
    x = tf.keras.layers.Reshape((shape[1] * shape[2], embed_dim))(x)
    return x, shape

def build_swin_denoiser():
    inputs = tf.keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
    x, shape = patch_embedding(inputs)
    x = swin_block(x)
    x = swin_block(x)
    h = shape[1]
    w = shape[2]
    c = shape[3]
    x = tf.keras.layers.Reshape((h, w, c))(x)
    x = tf.keras.layers.Conv2DTranspose(64,kernel_size=4,strides=4,padding="same")(x)
    outputs = tf.keras.layers.Conv2D(3,kernel_size=3,padding="same",activation="sigmoid")(x)
    model = tf.keras.Model(inputs, outputs)
    return model

model = build_swin_denoiser()

model.compile(
    optimizer="adam",
    loss="mse",
    metrics=["mae"]
)

model.summary()
history = model.fit(x_train,y_train,validation_data=(x_test, y_test),epochs=EPOCHS,batch_size=BATCH_SIZE)
denoised = model.predict(x_test)
```

```
Model: "functional"

┏━━━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┳━━━━━━━━━━━━┳━━━━━━━━━━━━━━━━━━━┓
┃ Layer (type)        ┃ Output Shape      ┃    Param # ┃ Connected to      ┃
┡━━━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━╇━━━━━━━━━━━━╇━━━━━━━━━━━━━━━━━━━┩
│ input_layer         │ (None, 128, 128,  │          0 │ -                 │
│ (InputLayer)        │ 3)                │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ conv2d (Conv2D)     │ (None, 32, 32,    │      3,136 │ input_layer[0][0] │
│                     │ 64)               │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ reshape (Reshape)   │ (None, 1024, 64)  │          0 │ conv2d[0][0]      │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ layer_normalization │ (None, 1024, 64)  │        128 │ reshape[0][0]     │
│ (LayerNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ multi_head_attenti… │ (None, 1024, 64)  │     66,368 │ layer_normalizat… │
│ (MultiHeadAttentio… │                   │            │ layer_normalizat… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ add (Add)           │ (None, 1024, 64)  │          0 │ reshape[0][0],    │
│                     │                   │            │ multi_head_atten… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ layer_normalizatio… │ (None, 1024, 64)  │        128 │ add[0][0]         │
│ (LayerNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense (Dense)       │ (None, 1024, 256) │     16,640 │ layer_normalizat… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense_1 (Dense)     │ (None, 1024, 64)  │     16,448 │ dense[0][0]       │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ add_1 (Add)         │ (None, 1024, 64)  │          0 │ add[0][0],        │
│                     │                   │            │ dense_1[0][0]     │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ layer_normalizatio… │ (None, 1024, 64)  │        128 │ add_1[0][0]       │
│ (LayerNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ multi_head_attenti… │ (None, 1024, 64)  │     66,368 │ layer_normalizat… │
│ (MultiHeadAttentio… │                   │            │ layer_normalizat… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ add_2 (Add)         │ (None, 1024, 64)  │          0 │ add_1[0][0],      │
│                     │                   │            │ multi_head_atten… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ layer_normalizatio… │ (None, 1024, 64)  │        128 │ add_2[0][0]       │
│ (LayerNormalizatio… │                   │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense_2 (Dense)     │ (None, 1024, 256) │     16,640 │ layer_normalizat… │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ dense_3 (Dense)     │ (None, 1024, 64)  │     16,448 │ dense_2[0][0]     │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ add_3 (Add)         │ (None, 1024, 64)  │          0 │ add_2[0][0],      │
│                     │                   │            │ dense_3[0][0]     │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ reshape_1 (Reshape) │ (None, 32, 32,    │          0 │ add_3[0][0]       │
│                     │ 64)               │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ conv2d_transpose    │ (None, 128, 128,  │     65,600 │ reshape_1[0][0]   │
│ (Conv2DTranspose)   │ 64)               │            │                   │
├─────────────────────┼───────────────────┼────────────┼───────────────────┤
│ conv2d_1 (Conv2D)   │ (None, 128, 128,  │      1,731 │ conv2d_transpose… │
│                     │ 3)                │            │                   │
└─────────────────────┴───────────────────┴────────────┴───────────────────┘

 Total params: 269,891 (1.03 MB)

 Trainable params: 269,891 (1.03 MB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 31s 2s/step - loss: 0.0422 - mae: 0.1585 - val_loss: 0.0071 - val_mae: 0.0618
Epoch 2/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0071 - mae: 0.0625 - val_loss: 0.0059 - val_mae: 0.0575
Epoch 3/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 18s 2s/step - loss: 0.0053 - mae: 0.0541 - val_loss: 0.0045 - val_mae: 0.0478
Epoch 4/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 19s 2s/step - loss: 0.0043 - mae: 0.0480 - val_loss: 0.0042 - val_mae: 0.0453
Epoch 5/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 20s 2s/step - loss: 0.0041 - mae: 0.0465 - val_loss: 0.0051 - val_mae: 0.0543
Epoch 6/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 19s 1s/step - loss: 0.0045 - mae: 0.0507 - val_loss: 0.0048 - val_mae: 0.0523
Epoch 7/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0040 - mae: 0.0470 - val_loss: 0.0037 - val_mae: 0.0426
Epoch 8/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0034 - mae: 0.0419 - val_loss: 0.0036 - val_mae: 0.0431
Epoch 9/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0039 - mae: 0.0468 - val_loss: 0.0033 - val_mae: 0.0412
Epoch 10/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0033 - mae: 0.0430 - val_loss: 0.0032 - val_mae: 0.0407
Epoch 11/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0030 - mae: 0.0399 - val_loss: 0.0027 - val_mae: 0.0362
Epoch 12/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 20s 1s/step - loss: 0.0025 - mae: 0.0354 - val_loss: 0.0022 - val_mae: 0.0328
Epoch 13/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0021 - mae: 0.0321 - val_loss: 0.0022 - val_mae: 0.0334
Epoch 14/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 22s 2s/step - loss: 0.0019 - mae: 0.0318 - val_loss: 0.0017 - val_mae: 0.0295
Epoch 15/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 16s 1s/step - loss: 0.0018 - mae: 0.0308 - val_loss: 0.0021 - val_mae: 0.0340
Epoch 16/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 16s 1s/step - loss: 0.0018 - mae: 0.0315 - val_loss: 0.0018 - val_mae: 0.0320
Epoch 17/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0017 - mae: 0.0305 - val_loss: 0.0015 - val_mae: 0.0279
Epoch 18/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0014 - mae: 0.0276 - val_loss: 0.0014 - val_mae: 0.0270
Epoch 19/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0014 - mae: 0.0271 - val_loss: 0.0013 - val_mae: 0.0261
Epoch 20/20
12/12 ━━━━━━━━━━━━━━━━━━━━ 17s 1s/step - loss: 0.0013 - mae: 0.0263 - val_loss: 0.0014 - val_mae: 0.0267
1/1 ━━━━━━━━━━━━━━━━━━━━ 2s 2s/step
```

In [11]:
```
plt.figure(figsize=(14, 10))

plt.subplot(1, 3, 1)
plt.imshow(x_test[0])
plt.title("Noisy")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(denoised[0])
plt.title("Denoised")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(y_test[0])
plt.title("Original")
plt.axis("off")

plt.show()
```

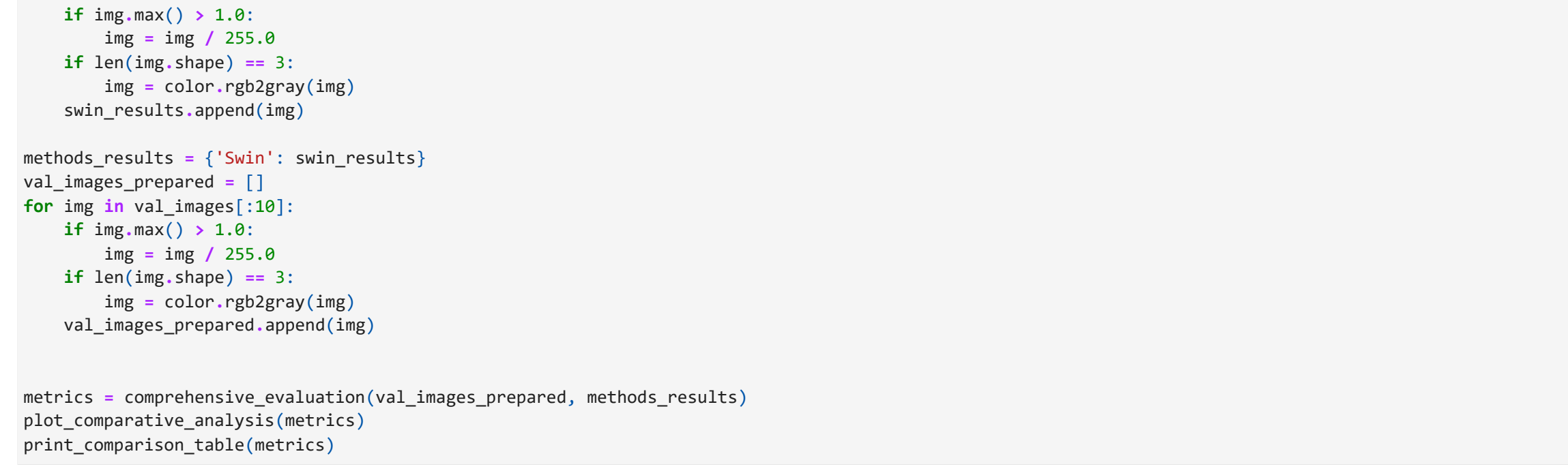


In [12]:
```
print(f"PSNR: {psnr(y_test[0], denoised[0], data_range=1.0):.4f}")
print(f"SSIM: {ssim(y_test[0],denoised[0],data_range=1.0,win_size=7, channel_axis=-1):.4f}")
print(f"Edge Preservation Index (EPI): {edge_preservation_index(y_test[0], denoised[0]):.4f}")
```

```
PSNR: 27.2142
SSIM: 0.7560
Edge Preservation Index (EPI): 0.9071
```

PHASE 4: Comparative and Critical Analysis

In [21]:
```
def comprehensive_evaluation(images, methods_results, noise_types=['gaussian', 'salt_pepper', 'speckle']):
    metrics = {'PSNR': {}, 'SSIM': {}, 'EPI': {}}

    for noise_type in noise_types:
        metrics['PSNR'][noise_type] = {}
        metrics['SSIM'][noise_type] = {}
        metrics['EPI'][noise_type] = {}

        for method in ['BM3D', 'CNN', 'Swin']:
            psnr_vals, ssim_vals, epi_vals = [], [], []

            for i, img in enumerate(images[:min(10, len(images))]):  # Test on first 10 images
                # Ensure image is in correct format
                if img.max() > 1.0:
                    img = img / 255.0

                # Generate noisy image
                if noise_type == 'gaussian':
                    noisy = add_gaussian_noise(img, sigma=0.1)
                elif noise_type == 'salt_pepper':
                    noisy = add_salt_pepper_noise(img)
                else:
                    noisy = add_speckle_noise(img)

                # Get denoised result
                if method == 'BM3D':
                    denoised = denoise_bm3d(noisy)
                elif method == 'CNN':
                    with torch.no_grad():
                        # Ensure noisy image is 2D for CNN
                        if len(noisy.shape) == 3:
                            noisy_gray = color.rgb2gray(noisy)
                        else:
                            noisy_gray = noisy
                        noisy_tensor = torch.FloatTensor(noisy_gray).unsqueeze(0).unsqueeze(0).to(device)
                        denoised = cnn_model(noisy_tensor).cpu().squeeze().numpy()
                else:  # Swin
                    if i < len(methods_results['Swin']):
                        denoised = methods_results['Swin'][i]
                        # Convert to grayscale if needed for fair comparison
                        if len(denoised.shape) == 3:
                            denoised = color.rgb2gray(denoised)
                    else:
                        denoised = img

                # Ensure denoised has same shape as original
                if denoised.shape != img.shape:
                    denoised = cv2.resize(denoised, (img.shape[1], img.shape[0]))

                # Calculate metrics
                psnr_vals.append(psnr(img, denoised, data_range=1.0))

                # Handle SSIM with proper win_size
                min_dim = min(img.shape[0], img.shape[1])
                win_size = min(7, min_dim if min_dim % 2 == 1 else min_dim - 1)
                if win_size < 3:
                    win_size = 3

                ssim_vals.append(ssim(img, denoised, data_range=1.0, win_size=win_size))
                epi_vals.append(edge_preservation_index(img, denoised))

            metrics['PSNR'][noise_type][method] = np.mean(psnr_vals) if psnr_vals else 0
            metrics['SSIM'][noise_type][method] = np.mean(ssim_vals) if ssim_vals else 0
            metrics['EPI'][noise_type][method] = np.mean(epi_vals) if epi_vals else 0

    return metrics

def plot_comparative_analysis(metrics):
    noise_types = list(metrics['PSNR'].keys())
    methods = ['BM3D', 'CNN', 'Swin']
    metrics_names = ['PSNR', 'SSIM', 'EPI']
    fig, axes = plt.subplots(1, 3, figsize=(18, 5))
    for idx, metric in enumerate(metrics_names):
        x = np.arange(len(noise_types))
        width = 0.25
        for i, method in enumerate(methods):
            values = [metrics[metric][noise][method] for noise in noise_types]
            axes[idx].bar(x + i*width, values, width, label=method, alpha=0.8)
        axes[idx].set_xlabel('Noise Type', fontsize=12)
        axes[idx].set_ylabel(metric, fontsize=12)
        axes[idx].set_title(f'{metric} Comparison Across Methods', fontsize=14, fontweight='bold')
        axes[idx].set_xticks(x + width)
        axes[idx].set_xticklabels([n.replace('_', ' ').title() for n in noise_types])
        axes[idx].legend(loc='best')
        axes[idx].grid(True, alpha=0.3)
        axes[idx].set_ylim([0, 1.1 if metric in ['SSIM', 'EPI'] else None])
    plt.tight_layout()
    plt.show()

def print_comparison_table(metrics):
    print("\n" + "="*90)
    print("EVALUATION  METRIC RESULTS FOR MEDICAL IMAGE DENOISING")
    print("="*90)

    for noise_type in metrics['PSNR'].keys():
        print(f"\n\033[1m{noise_type.upper()} NOISE:\033[0m")
        print("-"*70)
        print(f"{'Method':<12} {'PSNR (dB)':<15} {'SSIM':<15} {'EPI':<15}")
        print("-"*70)
        best_psnr = max(metrics['PSNR'][noise_type].items(), key=lambda x: x[1])
        best_ssim = max(metrics['SSIM'][noise_type].items(), key=lambda x: x[1])
        best_epi = max(metrics['EPI'][noise_type].items(), key=lambda x: x[1])

        for method in ['BM3D', 'CNN', 'Swin']:
            psnr_val = metrics['PSNR'][noise_type][method]
            ssim_val = metrics['SSIM'][noise_type][method]
            epi_val = metrics['EPI'][noise_type][method]

            # Add markers for best performance
            psnr_str = f"{psnr_val:<15.4f}"
            ssim_str = f"{ssim_val:<15.4f}"
            epi_str = f"{epi_val:<15.4f}"

            if method == best_psnr[0]:
                psnr_str += " ✓"
            if method == best_ssim[0]:
                ssim_str += " ✓"
            if method == best_epi[0]:
                epi_str += " ✓"

            print(f"{method:<12} {psnr_str} {ssim_str} {epi_str}")




swin_results = []
for i in range(min(10, len(denoised))):
    img = denoised[i]
    # Ensure consistent format
    if img.max() > 1.0:
        img = img / 255.0
    if len(img.shape) == 3:
        img = color.rgb2gray(img)
    swin_results.append(img)

methods_results = {'Swin': swin_results}
val_images_prepared = []
for img in val_images[:10]:
    if img.max() > 1.0:
        img = img / 255.0
    if len(img.shape) == 3:
        img = color.rgb2gray(img)
    val_images_prepared.append(img)


metrics = comprehensive_evaluation(val_images_prepared, methods_results)
plot_comparative_analysis(metrics)
print_comparison_table(metrics)
```



```
==========================================================================================
EVALUATION  METRIC RESULTS FOR MEDICAL IMAGE DENOISING
==========================================================================================

GAUSSIAN NOISE:
----------------------------------------------------------------------
Method       PSNR (dB)       SSIM            EPI
----------------------------------------------------------------------
BM3D         32.5094         ✓ 0.8359          ✓ 0.9539           ✓
CNN          26.1331         0.5218          0.8269
Swin         28.3263         0.7652          0.8413

SALT_PEPPER NOISE:
----------------------------------------------------------------------
Method       PSNR (dB)       SSIM            EPI
----------------------------------------------------------------------
BM3D         20.0520         0.3631          0.6449
CNN          20.7116         0.2866          0.6319
Swin         28.3263         ✓ 0.7652          ✓ 0.8413           ✓

SPECKLE NOISE:
----------------------------------------------------------------------
Method       PSNR (dB)       SSIM            EPI
----------------------------------------------------------------------
BM3D         20.4562         0.4908          0.6453
CNN          22.2620         0.4140          0.6876
Swin         28.3263         ✓ 0.7652          ✓ 0.8413           ✓
```